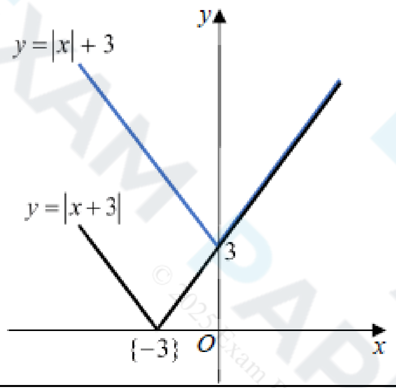
Mark Scheme

Q1.

<table>
<tr><th>Question</th><th colspan="2">Scheme</th><th>Marks</th><th>AOs</th></tr>
<tr><td></td><td colspan="2">Statement: "If m and n are irrational numbers, where $m \neq n$, then mn is also irrational."</td><td></td><td></td></tr>
<tr><td rowspan="3">(a)</td><td colspan="2">E.g. $m = \sqrt{3},\ n = \sqrt{12}$</td><td>M1</td><td>1.1b</td></tr>
<tr><td colspan="2">$\{mn =\}\ \left(\sqrt{3}\right)\left(\sqrt{12}\right) = 6$
$\Rightarrow$ statement untrue or 6 is not irrational or 6 is rational</td><td>A1</td><td>2.4</td></tr>
<tr><td colspan="2"></td><td>(2)</td><td></td></tr>
<tr><td rowspan="2">(b)(i), (ii) Way 1</td><td rowspan="2">
</td><td>V shaped graph {reasonably} symmetrical about the y-axis with vertical interpret $(0, 3)$ or 3 stated or marked on the positive y-axis</td><td>B1</td><td>1.1b</td></tr>
<tr><td>Superimposes the graph of $y = |x + 3|$ on top of the graph of $y = |x| + 3$</td><td>M1</td><td>3.1a</td></tr>
<tr><td rowspan="2"></td><td colspan="2">the graph of $y = |x| + 3$ is either the same or above the graph of $y = |x + 3|$ {for corresponding values of x}
or when $x \geq 0$, both graphs are equal (or the same)
when $x < 0$, the graph of $y = |x| + 3$ is above the graph of $y = |x + 3|$</td><td>A1</td><td>2.4</td></tr>
<tr><td colspan="2"></td><td>(3)</td><td></td></tr>
<tr><td rowspan="2">(b)(ii) Way 2</td><td rowspan="2"><u>Reason 1</u>
When $x \geq 0$, $|x| + 3 = |x + 3|$
<u>Reason 2</u>
When $x < 0$, $|x| + 3 > |x + 3|$</td><td>Any one of Reason 1 or Reason 2</td><td>M1</td><td>3.1a</td></tr>
<tr><td>Both Reason 1 and Reason 2</td><td>A1</td><td>2.4</td></tr>
<tr><td colspan="5">(5 marks)</td></tr>
</table>

Notes for Question	
(a)	
M1:	States or uses any pair of ***different*** numbers that will disprove the statement. E.g. $\sqrt{3},\ \sqrt{12}$; $\sqrt{2},\ \sqrt{8}$; $\sqrt{5}, -\sqrt{5}$; $\frac{1}{\pi}, 2\pi$; $3\mathrm{e}, \frac{4}{5\mathrm{e}}$;
A1:	Uses correct reasoning to disprove the given statement, with a correct conclusion
Note:	Writing $(3\mathrm{e})\left(\frac{4}{5\mathrm{e}}\right) = \frac{12}{5} \Rightarrow$ untrue is sufficient for M1A1
(b)(i)	
B1:	See scheme
(b)(ii)	
M1:	For constructing a method of comparing $\lvert x\rvert + 3$ with $\lvert x+3\rvert$. See scheme.
A1:	Explains fully why $\lvert x\rvert + 3 \geq \lvert x+3\rvert$. See scheme.
Note:	Do not allow either $x > 0,\ \lvert x\rvert + 3 \geq \lvert x+3\rvert$ or $x \geq 0,\ \lvert x\rvert + 3 \geq \lvert x+3\rvert$ as a valid reason
Note	$x = 0$ (or where necessary $x = -3$) need to be considered in their solutions for A1
Note:	Do not allow an incorrect statement such as $x \leq 0,\ \lvert x\rvert + 3 > \lvert x+3\rvert$ for A1
(b)(ii)	
Note:	Allow M1A1 for $x > 0,\ \lvert x\rvert + 3 = \lvert x+3\rvert$ and for $x \leq 0,\ \lvert x\rvert + 3 \geq \lvert x+3\rvert \geqslant$
Note:	Allow M1 for any of • x is positive, $\lvert x\rvert + 3 = \lvert x+3\rvert$ • x is negative, $\lvert x\rvert + 3 > \lvert x+3\rvert$ • $x > 0,\ \lvert x\rvert + 3 = \lvert x+3\rvert$ • $x \leq 0,\ \lvert x\rvert + 3 \geq \lvert x+3\rvert$ • $x > 0,\ \lvert x\rvert + 3$ and $\lvert x+3\rvert$ are equal • $x \geq 0,\ \lvert x\rvert + 3$ and $\lvert x+3\rvert$ are equal • when $x \geq 0$, both graphs are equal • for positive values $\lvert x\rvert + 3$ and $\lvert x+3\rvert$ are the same Condone for M1 • $x \leq 0,\ \lvert x\rvert + 3 > \lvert x+3\rvert$ • $x < 0,\ \lvert x\rvert + 3 \geq \lvert x+3\rvert$

(b)(ii) Way 3	• For $x > 0,\ \lvert x\rvert + 3 = \lvert x+3\rvert$ • For $-3 < x < 0$, as $\lvert x\rvert + 3 > 3$ and $\{0 <\}\ \lvert x+3\rvert < 3$, then $\lvert x\rvert + 3 > \lvert x+3\rvert$	M1	3.1a
	• For $x \leq -3$, as $\lvert x\rvert + 3 = -x + 3$ and $\lvert x+3\rvert = -x - 3$, then $\lvert x\rvert + 3 > \lvert x+3\rvert$	A1	2.4

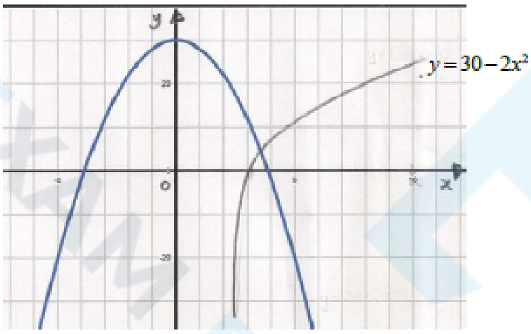
Q2.

Question	Scheme		Marks	AOs
(a)	$\frac{dr}{dt} \propto \pm\frac{1}{r^2}$ or $\frac{dr}{dt} = \pm\frac{k}{r^2}$ (for k or a numerical k)		M1	3.3
	$\int r^2\,dr = \int \pm k\,dt \Rightarrow \ldots$ (for k or a numerical k)		M1	2.1
	$\frac{1}{3}r^3 = \pm kt\ \{+c\}$		A1	1.1b
	$t=0,\ r=5$ and $t=4,\ r=3$	$t=0,\ r=5$ and $t=240,\ r=3$	M1	3.1a
	gives $\frac{1}{3}r^3 = -\frac{49}{6}t + \frac{125}{3}$, where r, in mm, is the radius {of the mint} and t, in minutes, is the time from when it {the mint} was placed in the mouth	gives $\frac{1}{3}r^3 = -\frac{49}{360}t + \frac{125}{3}$, where r, in mm, is the radius {of the mint} and t, in seconds, is the time from when it {the mint} was placed in the mouth	A1	1.1b
			(5)	
(b)	$r = 0 \Rightarrow 0 = -\frac{49}{6}t + \frac{125}{3} \Rightarrow\ 0 = -49t + 250\ \Rightarrow t = \ldots$		M1	3.4
	time = 5 minutes 6 seconds		A1	1.1b
			(2)	
(c)	Suggests a suitable limitation of the model. E.g. • Model does not consider how the mint is sucked • Model does not consider whether the mint is bitten • Model is limited for times up to 5 minutes 6 seconds, o.e. • Not valid for times greater than 5 minutes 6 seconds, o.e. • Mint may not retain the shape of a sphere (or have uniform radius) as it is being sucked • The model indicates that the radius of the mint is negative after it dissolves • Model does not consider the temperature in the mouth • Model does not consider rate of saliva production • Mint could be swallowed before it dissolves in the mouth		B1	3.5b
			(1)	
				(8 marks)

Notes for Question	
(a)	
M1:	Translates the description of the model into mathematics. See scheme.
M1:	Separates the variables of their differential equation which is in the form $\frac{\mathrm{d}r}{\mathrm{d}t} = \mathrm{f}(r)$ and some attempt at integration. (e.g. attempts to integrate at least one side). e.g. $\int r^2\,\mathrm{d}r = \int \pm k\,\mathrm{d}t$ and some attempt at integration. Condone the lack of integral signs
Note:	You can imply the M1 mark for $r^2 dr = -k\,\mathrm{d}t \Rightarrow \frac{1}{3}r^3 = -kt$
Note:	A numerical value of k (e.g. $k = \pm 1$) is allowed for the first two M marks
A1:	Correct integration to give $\frac{1}{3}r^3 = \pm kt$ with or without a constant of integration, c
M1:	For a complete process of using the boundary conditions to find both their unknown constants and finds an equation linking r and t So applies either • $t = 0,\ r = 5$ and $t = 4,\ r = 3$, or • $t = 0,\ r = 5$ and $t = 240,\ r = 3$, ***on their integrated equation*** to find their constants k and c ***and*** obtains an equation linking r and t
A1:	Correct equation, with variables r and t fully defined including correct reference to units. • $\frac{1}{3}r^3 = -\frac{49}{6}t + \frac{125}{3}$, {or an equivalent equation,} where r, in mm, is the radius {of the mint} and t, in minutes, is the time from when it {the mint} was placed in the mouth • $\frac{1}{3}r^3 = -\frac{49}{360}t + \frac{125}{3}$, {or an equivalent equation,} where r, in mm, is the radius {of the mint} and t, in seconds, is the time from when it {the mint} was placed in the mouth
Note:	Allow correct equations such as • in minutes, $r = \sqrt[3]{\frac{250-49t}{2}}$, $r^3 = -\frac{49}{2}t + 125$ or $t = \frac{250-2r^3}{49}$ • in seconds, $r = \sqrt[3]{\frac{15000-49t}{120}}$, $r^3 = -\frac{49}{120}t + 125$ or $t = \frac{15000-120r^3}{49}$
Note:	t defined as "the time from the start" is not sufficient for the final A1

(b)	
M1:	Sets $r = 0$ in their part (a) equation which links r with t and rearranges to make $t = \ldots$
A1:	5 minutes 6 seconds cao (**Note:** 306 seconds with no reference to 5 minutes 6 seconds is A0)
Note:	Give M0 if their equation would solve to give a negative time or a negative time is found
Note:	You can mark part (a) and part (b) together
(c)	
B1:	See scheme
Note:	Do not accept by itself • mint may not dissolve at a constant rate • rate of decrease of mint must be constant • $0 \le t < \frac{250}{49}$, $r \ge 0$; without any written explanation • reference to a mint having $r > 5$

Q3.

Question	Scheme	Marks	AOs
(a)	$f(3.5) = -4.8,\ f(4) = (+)3.1$	M1	1.1b
	Change of sign and function continuous in interval $[3.5, 4] \Rightarrow$ Root *	A1*	2.4
		(2)	
(b)	Attempts $x_1 = x_0 - \dfrac{f(x_0)}{f'(x_0)} \Rightarrow \quad x_1 = 4 - \dfrac{3.099}{16.67}$	M1	1.1b
	$x_1 = 3.81$	A1	1.1b
	$y = \ln(2x-5)$	(2)	
(c)	 Attempts to sketch both $y = \ln(2x-5)$ and $y = 30 - 2x^2$	M1	3.1a
	States that $y = \ln(2x-5)$ meets $y = 30 - 2x^2$ in just one place, therefore $y = \ln(2x-5) = 30 - 2x$ has just one root $\Rightarrow f(x) = 0$ has just one root	A1	2.4
		(2)	
			(6 marks)

Notes:
(a) **M1:** Attempts f(x) at both $x = 3.5$ and $x = 4$ with at least one correct to 1 significant figure **A1*:** f(3.5) and f(4) correct to 1 sig figure (rounded or truncated) with a correct reason and conclusion. A reason could be change of sign, or $f(3.5) \times f(4) < 0$ or similar with f(x) being continuous in this interval. A conclusion could be 'Hence root' or 'Therefore root in interval'
(b) **M1:** Attempts $x_1 = x_0 - \dfrac{f(x_0)}{f'(x_0)}$ evidenced by $x_1 = 4 - \dfrac{3.099}{16.67}$ **A1:** Correct answer only $x_1 = 3.81$
(c) **M1:** For a valid attempt at showing that there is only one root. This can be achieved by • Sketching graphs of $y = \ln(2x-5)$ and $y = 30 - 2x^2$ on the same axes • Showing that $f(x) = \ln(2x-5) + 2x^2 - 30$ has no turning points • Sketching a graph of $f(x) = \ln(2x-5) + 2x^2 - 30$ **A1:** Scored for correct conclusion

Q4.

Question	Scheme	Marks	AOs
(a)	$f(x) \geqslant 5$	B1	1.1b
		(1)	
(b)	Uses $-2(3-x)+5=\frac{1}{2}x+30$	M1	3.1a
	Attempts to solve by multiplying out bracket, collect terms etc $\frac{3}{2}x=31$	M1	1.1b
	$x=\frac{62}{3}$ only	A1	1.1b
		(3)	
(c)	Makes the connection that there must be two intersections. Implied by either end point $k>5$ or $k \leqslant 11$	M1	2.2a
	$\{k : k \in \mathbb{R}, 5<k \leqslant 11\}$	A1	2.5
		(2)	
			(6 marks)

Notes:

(a)

B1: $f(x) \geqslant 5$ Also allow $f(x) \in [5, \infty)$

(b)

M1: Deduces that the solution to $f(x)=\frac{1}{2}x+30$ can be found by solving

$-2(3-x)+5=\frac{1}{2}x+30$

M1: Correct method used to solve their equation. Multiplies out bracket/ collects like terms

A1: $x=\frac{62}{3}$ only. Do not allow 20.6

(c)

M1: Deduces that two distinct roots occurs when $y=k$ intersects $y=f(x)$ in two places. This may be implied by the sight of either end point. Score for sight of either $k>5$ or $k \leqslant 11$

A1: Correct solution only $\{k : k \in \mathbb{R}, 5<k \leqslant 11\}$

Q5.

Question	Scheme	Marks	AOs
(a)	Either attempts $\frac{3x-7}{x-2}=7 \Rightarrow x=...$ Or attempts $f^{-1}(x)$ and substitutes in $x=7$	M1	3.1a
	$\frac{7}{4}$ oe	A1	1.1b
		(2)	
(b)	Attempts $ff(x)=\frac{3\times\left(\frac{3x-7}{x-2}\right)-7}{\left(\frac{3x-7}{x-2}\right)-2}=\frac{3\times(3x-7)-7(x-2)}{3x-7-2(x-2)}$	M1, dM1	1.1b 1.1b
	$=\frac{2x-7}{x-3}$	A1	2.1
		(3)	
	(5 marks)		
Notes:			

(a)

M1: For either attempting to solve $\frac{3x-7}{x-2}=7$. Look for an attempt to multiply by the $(x-2)$ leading to a value for x.

Or score for substituting in $x=7$ in $f^{-1}(x)$. FYI $f^{-1}(x)=\frac{2x-7}{x-3}$

The method for finding $f^{-1}(x)$ should be sound, but you can condone slips.

A1: $\frac{7}{4}$

(b)

M1: For an attempt at fully substituting $\frac{3x-7}{x-2}$ into $f(x)$. Condone slips but the expression must have a correct form. E.g. $\frac{3\times\left(\frac{*-*}{*-*}\right)-a}{\left(\frac{*-*}{*-*}\right)-b}$ where a and b are positive constants.

dM1: Attempts to multiply **all** terms on the numerator and denominator by $(x-2)$ to create a fraction $\frac{P(x)}{Q(x)}$ where both $P(x)$ and $Q(x)$ are linear expressions. Condone $\frac{P(x)}{Q(x)}\times\frac{x-2}{x-2}$

A1: Reaches $\frac{2x-7}{x-3}$ via careful and accurate work. Implied by $a=2, b=-7$ following correct work.

....

Methods involving $\frac{3x-7}{x-2}\equiv a+\frac{b}{x-2}$ may be seen. The scheme can be applied in a similar way

FYI $\frac{3x-7}{x-2}\equiv 3-\frac{1}{x-2}$

Q6.

Question	Scheme	Marks	AOs
(a)	$\ln x \to \frac{1}{x}$	B1	1.1a
	Method to differentiate $\frac{4x^2+x}{2\sqrt{x}}$ – see notes	M1	1.1b
	E.g. $2\times\frac{3}{2}x^{\frac{1}{2}}+\frac{1}{2}\times\frac{1}{2}x^{-\frac{1}{2}}$	A1	1.1b
	$\frac{dy}{dx}=3\sqrt{x}+\frac{1}{4\sqrt{x}}-\frac{4}{x}=\frac{12x^2+x-16\sqrt{x}}{4x\sqrt{x}}$ *	A1*	2.1
		(4)	
(b)	$12x^2+x-16\sqrt{x}=0 \Rightarrow 12x^{\frac{3}{2}}+x^{\frac{1}{2}}-16=0$	M1	1.1b
	E.g. $12x^{\frac{3}{2}}=16-\sqrt{x}$	dM1	1.1b
	$x^{\frac{3}{2}}=\frac{4}{3}-\frac{\sqrt{x}}{12} \Rightarrow x=\left(\frac{4}{3}-\frac{\sqrt{x}}{12}\right)^{\frac{2}{3}}$ *	A1*	2.1
		(3)	
(c)	$x_2=\sqrt[3]{\left(\frac{4}{3}-\frac{\sqrt{2}}{12}\right)^2}$	M1	1.1b
	$x_2=$ awrt 1.13894	A1	1.1b
	$x=1.15650$	A1	2.2a
		(3)	
			(10 marks)

Notes:

(a)

B1: Differentiates $\ln x \to \frac{1}{x}$ seen or implied

M1: Correct method to differentiate $\frac{4x^2+x}{2\sqrt{x}}$:

Look for $\frac{4x^2+x}{2\sqrt{x}} \to ...x^{\frac{3}{2}}+...x^{\frac{1}{2}}$ being then differentiated to $Px^{\frac{1}{2}}+...$ or $...+Qx^{-\frac{1}{2}}$

Alternatively uses the quotient rule on $\frac{4x^2+x}{2\sqrt{x}}$.

Condone slips but if rule is not quoted expect $\left(\frac{dy}{dx}\right)=\frac{2\sqrt{x}(Ax+B)-(4x^2+x)Cx^{-\frac{1}{2}}}{(2\sqrt{x})^2}$ $(A,B,C>0)$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu'-uv'}{v^2}$ etc.

Alternatively uses the product rule on $(4x^2+x)(2\sqrt{x})^{-1}$

Condone slips but expect $\left(\frac{dy}{dx}\right)=Ax^{-\frac{1}{2}}(Bx+C)+D(4x^2+x)x^{-\frac{3}{2}}$ $(A,B,C>0)$

In general condone missing brackets for the M mark. If they quote $u=4x^2+x$ and $v=2\sqrt{x}$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: Correct differentiation of $\frac{4x^2+x}{2\sqrt{x}}$ although may not be simplified.

Examples: $\left(\frac{dy}{dx}\right)=\frac{2\sqrt{x}(8x+1)-(4x^2+x)x^{-\frac{1}{2}}}{(2\sqrt{x})^2}$, $\frac{1}{2}x^{-\frac{1}{2}}(8x+1)-\frac{1}{4}(4x^2+x)x^{-\frac{3}{2}}$, $2\times\frac{3}{2}x^{\frac{1}{2}}+\frac{1}{2}\times\frac{1}{2}x^{-\frac{1}{2}}$

A1*: Obtains $\frac{dy}{dx}=\frac{12x^2+x-16\sqrt{x}}{4x\sqrt{x}}$ via $3\sqrt{x}+\frac{1}{4\sqrt{x}}-\frac{4}{x}$ or a correct application of the quotient or product rule and with sufficient working shown to reach the printed answer.
There must be no errors e.g. missing brackets.

(b)

M1: Sets $12x^2+x-16\sqrt{x}=0$ and divides by $\sqrt{x}$ or equivalent e.g. divides by x and multiplies by $\sqrt{x}$

dM1: Makes the term in $x^{\frac{3}{2}}$ the subject of the formula
A1*: A correct and rigorous argument leading to the given solution.

Alternative - working backwards:

$$x=\left(\frac{4}{3}-\frac{\sqrt{x}}{12}\right)^{\frac{2}{3}}\Rightarrow x^{\frac{3}{2}}=\frac{4}{3}-\frac{\sqrt{x}}{12}\Rightarrow 12x^{\frac{3}{2}}=16-\sqrt{x}\Rightarrow 12x^2=16\sqrt{x}-x\Rightarrow 12x^2-16\sqrt{x}+x=0$$

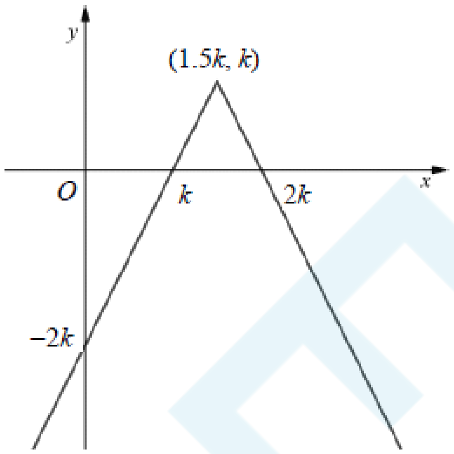
M1: For raising to power of 3/2 both sides. dM1: Multiplies through by $\sqrt{x}$. A1: Achieves printed answer and makes a minimal comment e.g. tick, #, QED, true etc.

(c)

M1: Attempts to use the iterative formula with $x_1=2$. This is implied by sight of $x_2=\left(\frac{4}{3}-\frac{\sqrt{2}}{12}\right)^{\frac{2}{3}}$ or awrt 1.14

A1: $x_2=$ awrt 1.13894
A1: Deduces that $x=1.15650$

Q7.

Question	Scheme	Marks	AOs
(a)			
	$\wedge$ shape in any position	B1	1.1b
	Correct x-intercepts or coordinates	B1	1.1b
	Correct y-intercept or coordinates	B1	1.1b
	Correct coordinates for the vertex of a $\wedge$ shape	B1	1.1b
		(4)	
(b)	$x = k$	B1	2.2a
	$k-(2x-3k)=x-k \Rightarrow x = ...$	M1	3.1a
	$x=\frac{5k}{3}$	A1	1.1b
	Set notation is required here for this mark $\left\{x : x < \frac{5k}{3}\right\} \cap \{x : x > k\}$	A1	2.5
		(4)	
(c)	$x = 3k$ **or** $y = 3 - 5k$	B1ft	2.2a
	$x = 3k$ **and** $y = 3 - 5k$	B1ft	2.2a
		(2)	
			(10 marks)

Notes
(a) **Note that the sketch may be seen on Figure 4** B1: See scheme B1: Correct x-intercepts. Allow as shown or written as $(k, 0)$ and $(2k, 0)$ and condone coordinates written as $(0, k)$ and $(0, 2k)$ as long as they are in the correct places. B1: Correct y-intercept. Allow as shown or written as $(0, -2k)$ or $(-2k, 0)$ as long as it is in the correct place. Condone $k - 3k$ for $-2k$. B1: Correct coordinates as shown **Note that the marks for the intercepts and the maximum can be seen away from the sketch but the coordinates must be the right way round or e.g. as $y = 0, x = k$ etc. These marks can be awarded without a sketch but if there is a sketch, such points must not contradict the sketch.** (b) B1: Deduces the correct critical value of $x = k$. May be implied by e.g. $x > k$ or $x < k$ M1: Attempts to solve $k - (2x - 3k) = x - k$ or an equivalent equation/inequality to find the other critical value. Allow this mark for reaching $k = \ldots$ or $x = \ldots$ as long as they are solving the required equation. A1: Correct value A1: Correct answer using the correct set notation. Allow e.g. $\left\{x : x \in \mathbb{R}, k < x < \frac{5k}{3}\right\}$, $\left\{x :\ k < x < \frac{5k}{3}\right\}$, $x \in \left(k, \frac{5k}{3}\right)$ and allow "|" for ":" But $\left\{x : x < \frac{5k}{3}\right\} \cup \{x : x > k\}$ scores A0 $\left\{x :\ k < x,\ x < \frac{5k}{3}\right\}$ scores A0 (c) B1ft: Deduces one correct coordinate. Follow through their maximum coordinates from (a) so allow $x = 2\times$"$1.5k$" or $y = 3 - 5\times$"k" but must be in terms of k. Allow as coordinates or $x = .., y = \ldots$ B1ft: Deduces both correct coordinates. Follow through their maximum coordinates from (a) so allow $x = 2\times$"$1.5k$" and $y = 3 - 5\times$"k" but must be in terms of k. Allow as coordinates or $x = .., y = \ldots$ If coordinates are given the wrong way round and not seen correctly as $x = \ldots, y = \ldots$ e.g. $(3 - 5k, 3k)$ this is B0B0

Alternative to part (b) by squaring:

$$k - |2x - 3k| = x - k \Rightarrow |2x - 3k| = 2k - x$$

$$4x^2 - 12kx + 9k^2 = 4k^2 - 4kx + x^2 \Rightarrow 3x^2 - 8kx + 5k^2 = 0$$

$$(3x - 5k)(x - k) = 0 \Rightarrow x = \frac{5k}{3},\ k$$

Score M1 for isolating the $|2x - 3k|$, squaring both sides to obtain 3 appropriate terms for each side, collects terms to obtain $Ax^2 + Bkx + Ck^2 = 0$ and solves for x

A1 for $x = \frac{5k}{3}$ and B1 for $x = k$

Then A1 as in the scheme.

Q8.

Via firstly integrating

Question	Scheme	Marks	AOs
	$f'(x)=6x^2+ax-23 \Rightarrow f(x)=2x^3+\frac{1}{2}ax^2-23x+c$	M1 A1	1.1b 1.1b
	$"c"=-12$	B1	2.2a
	$f(-4)=0 \Rightarrow 2\times(-4)^3+\frac{1}{2}a(-4)^2-23(-4)-12=0$	dM1	3.1a
	$a=\ldots$ (6)	dM1	1.1b
	$(f(x)=)2x^3+3x^2-23x-12$ Or Equivalent e.g. $(f(x)=)(x+4)(2x^2-5x-3)$ $(f(x)=)(x+4)(2x+1)(x-3)$	A1cso	2.1
		(6)	
			(6 marks)

Notes:

M1: Integrates $f'(x)$ with two correct indices. There is no requirement for the $+c$

A1: Fully correct integration (may be unsimplified). The $+c$ must be seen (or implied by the -12)

B1: Deduces that the constant term is -12

dM1: Dependent upon having done some integration. It is for setting up a linear equation in a by using $f(-4)=0$
May also see long division attempted for this mark. Need to see a complete method leading to a remainder in terms of a which is then set $=0$.
For reference, the quotient is $2x^2+\left(\frac{a}{2}-8\right)x+9-2a$ and the remainder is $8a-48$
May also use $(x+4)(px^2+qx+r)=2x^3+\frac{1}{2}ax^2-23x-12$ and compare coefficients to find p, q and r and hence a. Allow this mark if they solve for p, q and r
Note that some candidates use 2f(x) which is acceptable and gives the same result if executed correctly.

dM1: Solves the linear equation in a or uses p, q and r to find a.
It is dependent upon having attempted some integration and used $f(\pm 4)=0$ or long division/comparing coefficients with $(x+4)$ as a factor.

A1cso: For $(f(x)=)2x^3+3x^2-23x-12$ oe. Note that "f(x) =" does not need to be seen and ignore any "= 0"

Via firstly using factor

Question	Scheme	Marks	AOs
Alt	$f(x)=(x+4)(Ax^2+Bx+C)$	M1 A1	1.1b 1.1b
	$f(x)=Ax^3+(4A+B)x^2+(4B+C)x+4C \Rightarrow C=-3$	B1	2.2a
	$f'(x)=3Ax^2+2(4A+B)x+(4B+C)$ and $f'(x)=6x^2+ax-23$ $\Rightarrow A=\ldots$	dM1	3.1a
	Full method to get A, B and C	dM1	1.1b
	$f(x)=(x+4)(2x^2-5x-3)$	A1cso	2.1
		(6)	
			(6 marks)

Notes:

M1: Uses the fact that f(x) is a cubic expression with a factor of $(x+4)$

A1: For $f(x)=(x+4)(Ax^2+Bx+C)$

B1: Deduces that $C=-3$

dM1: Attempts to differentiate either by product rule or via multiplication and compares to $f'(x) = 6x^2 + ax - 23$ to find A.
dM1: Full method to get A, B and C
A1cso: $f(x) = (x + 4)(2x^2 - 5x - 3)$ or $f(x) = (x + 4)(2x + 1)(x - 3)$

Q9.

Question	Scheme	Marks	AOs
(a)	$x^2+8x-3=(Ax+B)(x+2)+C$ or $Ax(x+2)+B(x+2)+C$ $\Rightarrow A=\ldots, B=\ldots, C=\ldots$ **or** $x+2\overline{)x^2+8x-3}$ giving quotient $x+6$: x^2+2x, $6x-3$, $6x+12$, -15	M1	1.1b
	Two of $A=1, B=6, C=-15$	A1	1.1b
	All three of $A=1, B=6, C=-15$	A1	1.1b
		(3)	
6(b)	$\int \frac{x^2+8x-3}{x+2}\,dx = \int x+6-\frac{15}{x+2}\,dx = \ldots -15\ln(x+2)$	M1	1.1b
	$=\frac{1}{2}x^2+6x-15\ln(x+2) \quad (+c)$	A1ft	1.1b
	$\int_0^6 \frac{x^2+8x-3}{x+2}\,dx = \left[\frac{1}{2}x^2+6x-15\ln(x+2)\right]_0^6$ $= (18 + 36 - 15\ln 8) - (0 + 0 - 15\ln 2)$ $= 18 + 36 - (15 - 45)\ln 2$ or e.g. $18+36+15\ln\left(\frac{2}{8}\right)$	M1	2.1
	$= 54 - 30\ln 2$	A1	1.1b
		(4)	
			(7 marks)

Notes:

(a)

M1: Multiplies by $(x + 2)$ and attempts to find values for A, B and C e.g. by comparing coefficients or substituting values for x. If the method is unclear, at least 2 terms must be correct on rhs.
Or attempts to divide $x^2 + 8x - 3$ by $x + 2$ and obtains a linear quotient and a constant remainder.
This mark may be implied by 2 correct values for A, B or C

A1: Two of $A = 1$, $B = 6$, $C = -15$. But note that <u>just</u> performing the division correctly is insufficient and they must clearly identify their A, B, C to score any accuracy marks.

A1: All three of $A = 1$, $B = 6$, $C = -15$
This is implied by stating $\frac{x^2+8x-3}{x+2} = x + 6 - \frac{15}{x+2}$ or within the integral in (b)

(b)

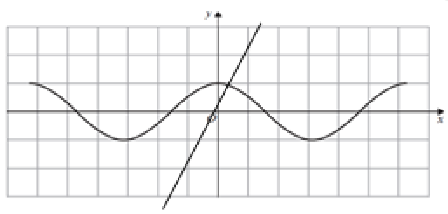
M1: Integrates an expression of the form $\frac{C}{x+2}$ to obtain $k\ln(x + 2)$.
Condone the omission of brackets around the "$x + 2$"

A1ft: Correct integration ft on their $Ax + B + \frac{C}{x+2}$, $(A, B, C \neq 0)$ The brackets should be present around the "$x + 2$" unless they are implied by subsequent work.

M1: Substitutes both limits 0 and 6 into an expression that contains an x or x^2 term or both and a ln term and subtracts either way round **WITH** fully correct log work to combine two log terms (but allow sign errors when removing brackets) leading to an answer of the form $a + b\ln c$ (a, b and c not necessarily integers)
e.g. if they expand to get $-15\ln 8 - 15\ln 2$ followed by $-15\ln 16$ and reach $a + b\ln c$ then allow the M mark

A1: $54 - 30\ln 2$ (Apply isw once a correct answer is seen)

Q10.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)		M1	This mark is given for plotting the line $y = 2x + \frac{1}{2}$ on the diagram with a correct gradient and intercept
	Only one intersection means that there is one root	A1	This mark is given for a reason why there is only one real root
(b)	$1 - \frac{x^2}{2} - 2x - \frac{1}{2} = 0$	M1	This mark is given for using the small angle approximation $\cos x = 1 - \frac{x^2}{2}$ in the given equation
	$x^2 + 4x - 1 = 0$	M1	This mark is igvne for rearranging to find a quadratic equation to solve
	0.236 or $-2 + \sqrt{5}$	A1	This mark is given for finding the correct (positive) solution for x

Q11.

Part	Working or answer an examiner might expect to see	Mark	Notes
(i)	For even numbers $n = 2m$, $n^2 + 2 = 4m^2 + 2$	M1	This mark is given for showing the case for all even numbers
	This is a multiple of 4 with 2 added, so cannot be divisible by 4	A1	This mark is given for a correct conclusion with a reason why $n^2 + 2$ is not divisible by 4 for all even numbers
	For odd numbers $n = 2m + 1$, $n^2 + 2 = (2m + 1)^2 + 2 = 4m^2 + 4m + 3$ $= 4(m^2 + m) + 3$	M1	This mark is given for showing the case for all odd numbers
	This is a multiple of 4 with 3 added, so cannot be divisible by 4 Hence, for all $n \in \mathbb{N}$, $n + 2$ is not divisible by 4	A1	This mark is given for a correct conclusion with a reason why $n^2 + 2$ is not divisible by 4 for all odd numbers and a full concluding statement that for all $n \in \mathbb{N}$, $n + 2$ is not divisible by 4
(ii)	For example, for $x = 9.4$ $\lvert 3x - 28 \rvert = 0.2$ and $(x - 9) = 0.4$	M1	This mark is given for showing that the statement is not true for $9.25 < x < 9.5$
	The statement is sometimes true; For example, for $x = 12$ $\lvert 3x - 28 \rvert = 8$ and $(x - 9) = 3$	A1	This mark is given for a correct statement and an example where the statement is true

Q12.

Question	Scheme	Marks	AOs
	For an attempt to solve **Either** $3-2x=7+x \Rightarrow x=\dots$ **or** $2x-3=7+x \Rightarrow x=\dots$	M1	1.1b
	Either $x=-\frac{4}{3}$ **or** $x=10$	A1	1.1b
	For an attempt to solve **Both** $3-2x=7+x \Rightarrow x=\dots$ **and** $2x-3=7+x \Rightarrow x=\dots$	dM1	1.1b
	For both $x=-\frac{4}{3}$ and $x=10$ with no extra solutions	A1	1.1b
		(4)	
ALT	**Alternative by squaring:**		
	$(3-2x)^2=(7+x)^2 \Rightarrow 9-12x+4x^2=49+14x+x^2$	M1	1.1b
	$3x^2-26x-40=0$	A1	1.1b
	$3x^2-26x-40=0 \Rightarrow x=\dots$	dM1	1.1b
	For both $x=-\frac{4}{3}$ and $x=10$ with no extra solutions	A1	1.1b
			(4 marks)
Notes:			

Note this question requires working to be shown not <u>just answers written down</u>. But correct equations seen followed by the correct answers can score full marks.

M1: Attempts to solve either correct equation.
Allow equivalent equations e.g. $3-2x=-7-x \Rightarrow x=...$

A1: One correct solution. Allow exact equivalents for $-\frac{4}{3}$ e.g. $-1\frac{1}{3}$ or $-1.\dot{3}$ but not e.g. -1.33

dM1: Attempts to solve both correct equations.
Allow equivalent equations e.g. $3-2x=-7-x \Rightarrow x=...$ **Depends on the first method mark.**

A1: For both $x=-\frac{4}{3}$ and $x=10$ with no extra solutions and neither clearly rejected but ignore any attempts to find the y coordinates whether correct or otherwise and ignore reference to e.g. $x = -7$ (from where $y = 7 + x$ intersects the x-axis) or $x=1.5$ (from finding the value of x at the vertex) as "extras". Allow exact equivalents for $-\frac{4}{3}$ e.g. $-1\frac{1}{3}$ or $-1.\dot{3}$ but not rounded e.g. -1.33

Isw if necessary e.g. ignore subsequent attempts to put the values in an inequality e.g. $-\frac{4}{3}<x<10$

But if e.g. $x=-\frac{4}{3}$ is obtained and a candidate states $x=\left|-\frac{4}{3}\right|$ then score A0

<u>Alternative solution via squaring</u>

M1: Attempts to square both sides. Condone poor squaring e.g. $(3-2x)^2=9\pm4x^2$ or $9\pm2x^2$

A1: Correct quadratic equation $3x^2-26x-40=0$. The "= 0" may be implied by their attempt to solve. Terms must be collected but not necessarily all on one side so allow e.g. $3x^2-26x=40$

dM1: Correct attempt to solve a **3 term** quadratic. See general guidance for solving a quadratic equation. The roots can be written down from a calculator so the method may be implied by their values. **Depends on the first method mark.**

A1: For both $x=-\frac{4}{3}$ and $x=10$ with no extra solutions and neither clearly rejected but ignore any attempts to find the y coordinates and do not count e.g. $x = -7$ (from where $y = 7 + x$ intersects the x-axis) or $x=1.5$ (from finding the value of x at the vertex) as "extras". Allow exact equivalents for $-\frac{4}{3}$ e.g. $-1\frac{1}{3}$ or $-1.\dot{3}$ but not e.g. -1.33

Isw if necessary e.g. ignore subsequent attempts to put the values in an inequality e.g. $-\frac{4}{3}<x<10$

But if e.g. $x=-\frac{4}{3}$ is obtained and a candidate states $x=\left|-\frac{4}{3}\right|$ then score A0